

Why Transport Maps Cannot Split Mass

A transport map cannot split mass because it is a deterministic function. If

$$T : X \rightarrow Y$$

is a map, then every source point x is sent to exactly one target point $T(x)$. Thus all mass located at x must move together to the single point $T(x)$.

In measure notation, the transported measure is

$$T_{\#}\mu(A) = \mu(T^{-1}(A)).$$

This formula has no mechanism for sending part of the mass at x to one point and another part to a different point.

Example

Let

$$\mu = \delta_0, \quad \nu = \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1.$$

No transport map T can satisfy $T_{\#}\mu = \nu$, because

$$T_{\#}\delta_0 = \delta_{T(0)}.$$

This is always a single Dirac mass, not a mixture of two Dirac masses.

A transport plan can split the mass:

$$\pi = \frac{1}{2}\delta_{(0,-1)} + \frac{1}{2}\delta_{(0,1)}.$$

This plan sends half the mass at 0 to -1 and half to 1 .